

Scoring Games

Next, we'll create a function that can keep score for a dots and boxes game. We start with a function that can take any given game and find the completed squares that have been drawn, and then we create a function that will calculate the score. Our function will count completed squares by iterating over every line segment in the game. If a line is a horizontal line, we determine whether it is the top of a completely drawn square by checking whether the parallel line below it has also been drawn in the game, and also whether the left and right sides of the square have been drawn. The function in [Listing 10-2](#) accomplishes this:

```
def squarefinder(game):
    countofsquares = 0
    for line in game:
        parallel = False
        left=False
        right=False
        if line[0][1]==line[1][1]:
            if [(line[0][0],line[0][1]-1),(line[1][0],line[1][1] - 1)] in game:
                parallel=True
            if [(line[0][0],line[0][1]), (line[1][0]-1,line[1][1] - 1)] in game:
                left=True
            if [(line[0][0]+1,line[0][1]), (line[1][0],line[1][1] - 1)] in game:
                right=True
            if parallel and left and right:
                countofsquares += 1
    return(countofsquares)
```

[Listing 10-2](#): A function that counts the number of squares that appear in a dots and boxes game board

You can see that the function returns the value of `countofsquares`, which we initialized with a 0 value at the beginning of the function. The function's `for` loop iterates over every line segment in a game. We start out assuming that neither the parallel line below this line nor the left and right lines that would connect these parallel lines have been played in the game so far. If a given line is a horizontal line, we check for the existence of those parallel, left, and right lines. If all four lines of the square we've checked are listed in the game, then we increment the `countofsquares` variable by 1. In this way,